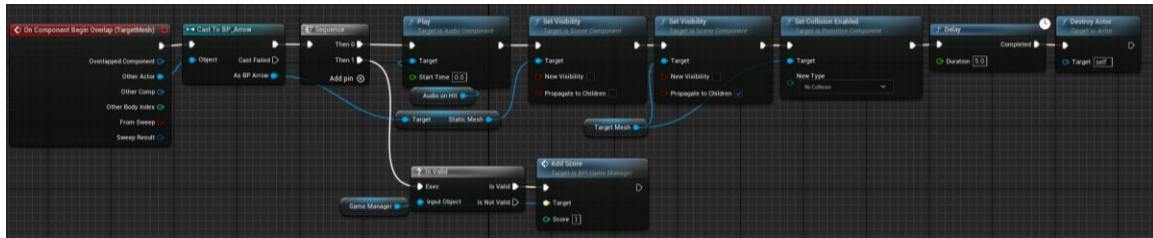


Purpose

At the instant the arrow touches the balloon, the full "explosion" feedback plays: play the sound, hide the model, disable collision to prevent re-triggering, stop the Tick ascent, destroy after a delay, and report the score to the GameManager.

Blueprint screenshots



Key points

- Disable collision before the delayed destruction: with collision off, other arrows can no longer trigger this balloon, preventing the same balloon from being re-popped and scored
- Hide on explosion rather than destroying immediately: the 5-second Delay lets the sound play fully; immediate destruction would cut the sound off
- Set Actor Tick Enabled(false): prevents Tick from running empty for 5 seconds after the explosion with the invisible balloon still computing its ascent; disabling Tick at the explosion moment saves the wasted computation

2. GameManager_Balloon (Balloon Manager)

2.1 Overview

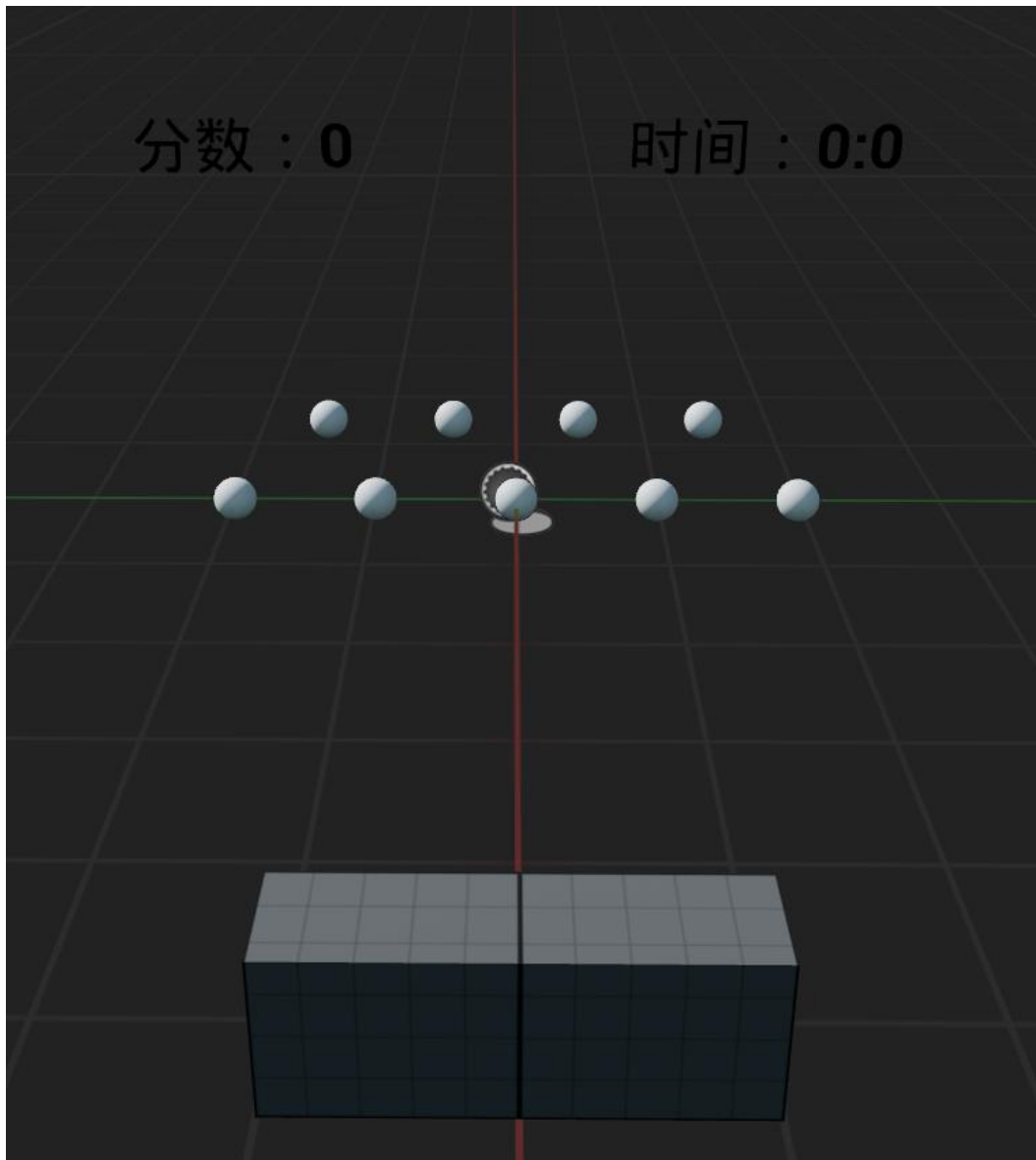
Purpose

The balloon shooting game's manager, sharing the same framework as the whack-a-mole and archery GameManagers. It configures the level parameters by difficulty, manages the 9 spawn points, loops balloon spawning at a fixed or random rhythm, and controls spawn start/stop through the dual check of game mode and game state.

Scene layout

9 spawn points (sphere models Sphere 1–9) are placed in the scene in two rows, front and back. The player stands in the middle shooting with the bow; balloons pop out of the spawn points and rise. Higher difficulty enables more spawn points (3 on Simple → 9 on Extreme).

Scene screenshots



Core variables

Variable	Type	Description
Create Balloon Positions	Scene Component array	The 9 spawn point position components; indices correspond one-to-one with Point Meshes
Point Meshes	Static Mesh array	The 9 spawn point sphere models, used for visualization
Level Data	Struct	The current difficulty's level parameters (interval / size / spawn point count, etc.)
Create Pos Index	Integer	The sequential mode's current index, cycling by

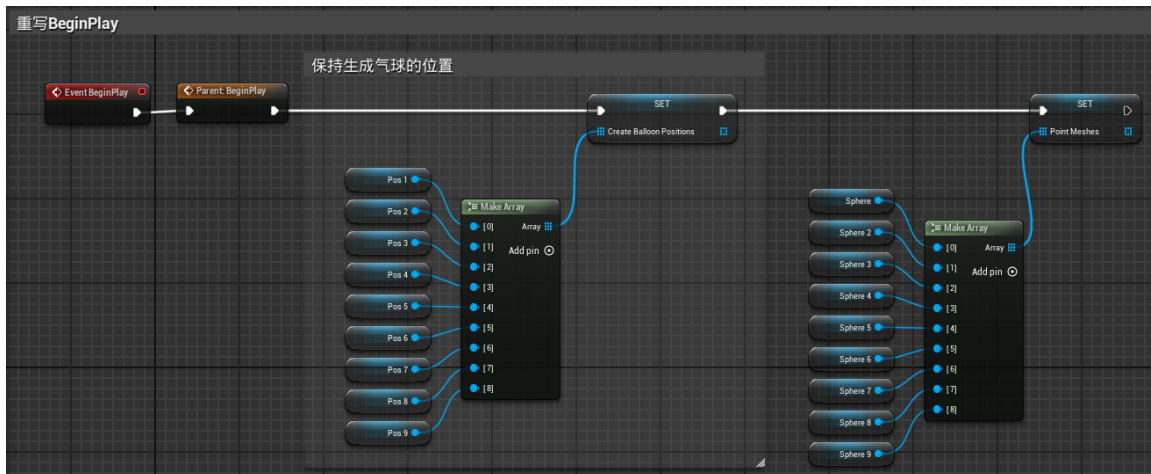
		modulo
Spawn Balloon Location	Vector	Cache of this spawn's position

2.2 BeginPlay — Initializing the Spawn Points

Purpose

Collects the 9 spawn point position components and sphere models placed in the editor into two parallel arrays for later use by difficulty. This is the spawning system's base data structure.

Blueprint screenshots



Key points

- The two arrays' indices correspond one-to-one: position[i] and sphere[i] are the same spawn point, with display control and position reading sharing the same index
- The position components carry the data and the sphere models carry the visualization: data and presentation are separated; the spheres are just markers for the player

2.3 SetLevelData — Difficulty Configuration

Purpose

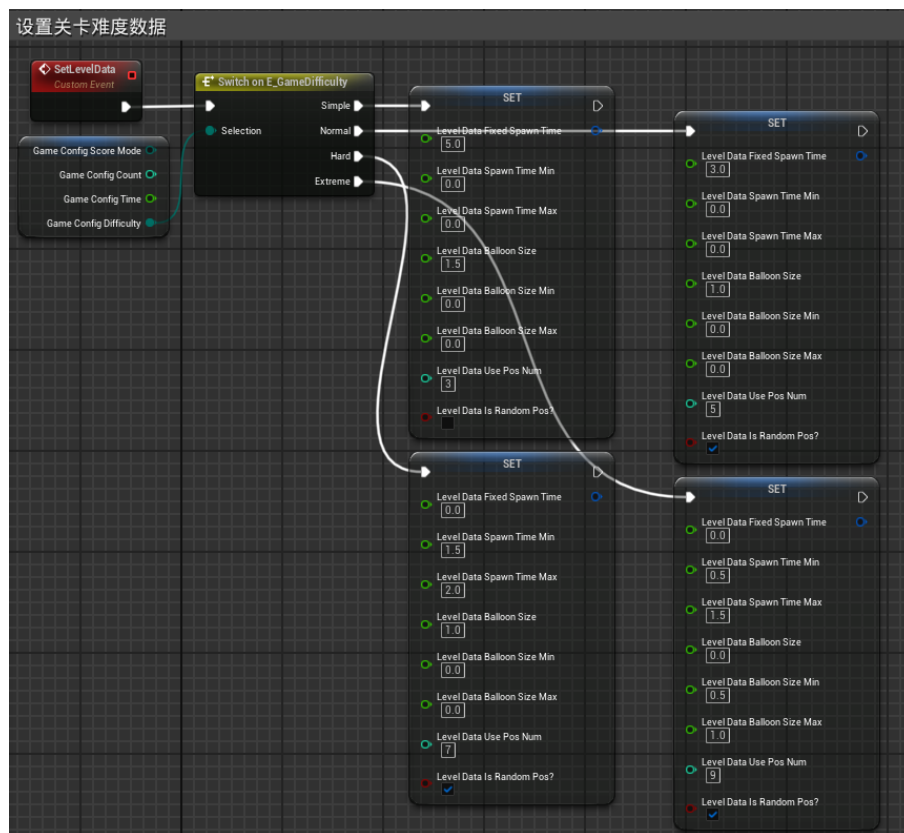
Per the difficulty enum in Game Config, writes the matching spawn interval, balloon size, spawn point count, and other parameters into the Level Data struct, driving all later spawning behavior. Difficulty is the source of the game's pressure.

Difficulty parameter table

Parameter	Simple	Normal	Hard	Extreme
Fixed Spawn Time (fixed interval)	5.0	3.0	0	0

Spawn Time Min / Max (random interval)	0 / 0	0 / 0	1.5 / 2.0	0.5 / 1.5
Balloon Size (fixed size)	1.5	1.0	1.0	0
Balloon Size Min / Max (random size)	0 / 0	0 / 0	0 / 0	0.5 / 1.0
Use Pos Num (enabled spawn points)	3	5	7	9
Is Random Pos? (random selection)	Yes	Yes	Yes	Yes

Blueprint screenshots



Key points

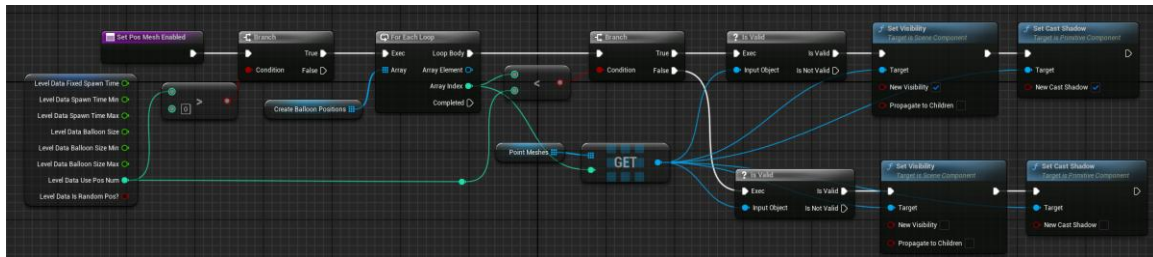
- 0 acts as the mode switch: Fixed Spawn Time=0 means the random-time mode, Balloon Size=0 means the random-size mode — no extra mode booleans needed
- Difficulty tightens on three dimensions at once: shorter intervals (fixed 5 s → random 0.5–1.5 s), smaller balloons (fixed 1.5 → random 0.5–1.0), and more spawn points (3 → 9)

2.4 SetPosMeshEnabled — Spawn Point Visualization

Purpose

Shows/hides the sphere models per the number of spawn points enabled by the current difficulty (Use Pos Num), letting the player see directly which positions will produce balloons.

Blueprint screenshots



Key points

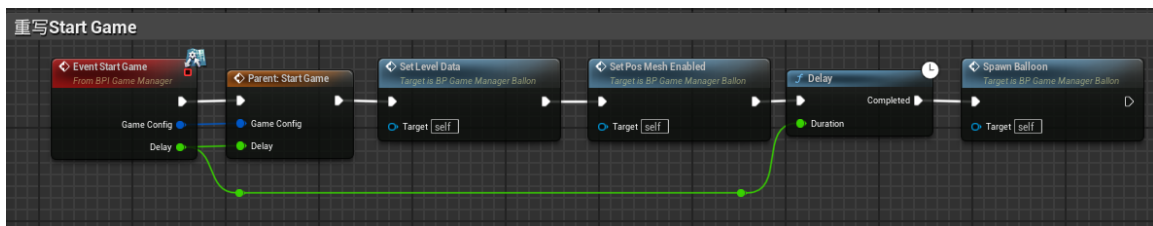
- What is shown is "the spawn points this difficulty enables": the player sees the balloon range from the start — a fairness design
- Casting is turned off when hidden too: no leftover shadows to break the illusion
- The spawn point spheres have no collision: purely visual markers that never block the balloons' spawning and rising

2.5 StartGame — Game Start

Purpose

Receives the game start event, completes the difficulty configuration and spawn point display, and after a delay spawns the first balloon, starting the spawn loop.

Blueprint screenshots



Key points

- Spawn Balloon is both "spawn one balloon" and the loop's starting point: after spawning it calls itself via a timer, forming a self-scheduling loop (see 2.6)
- The configure-then-start order cannot be reversed: without SetLevelData running, SpawnBalloon reads LevelData as all zeros

2.6 SpawnBloon — Main Spawn Loop

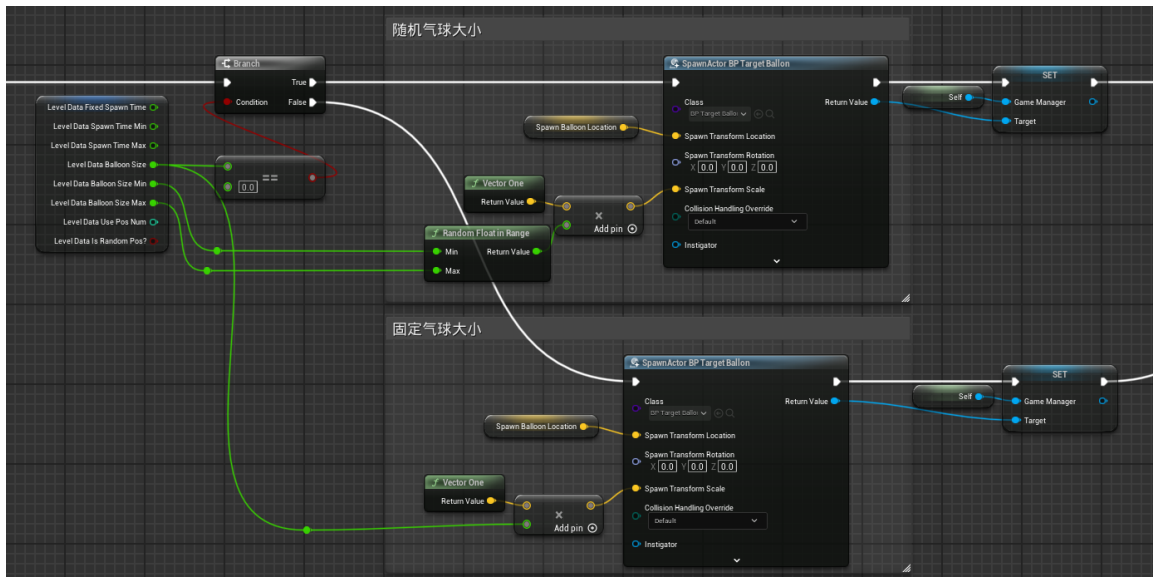
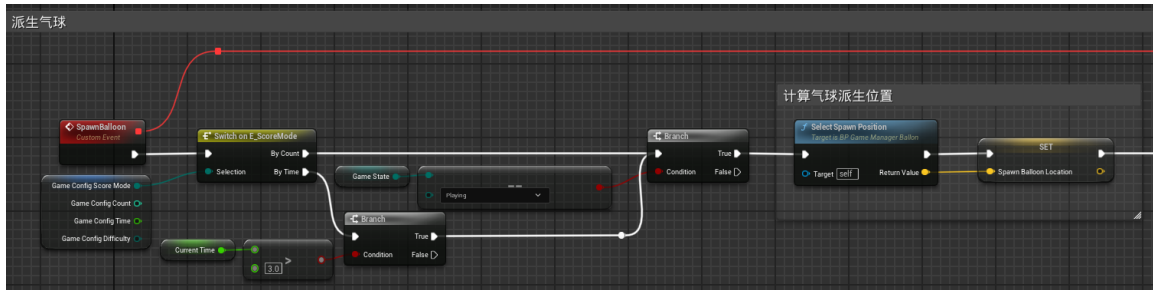
Purpose

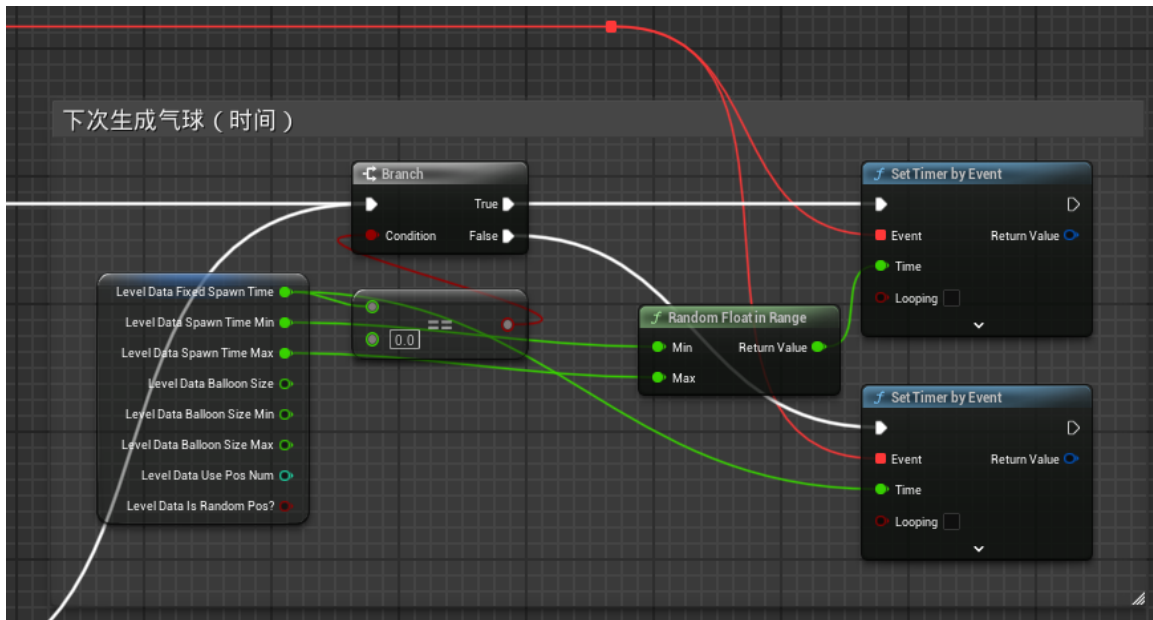
The manager's core event; each trigger runs one complete flow: check whether the game mode and state allow spawning → choose the spawn position → spawn the balloon at a fixed or random size → set the next spawn timer at a fixed or random interval.

State branch table

Game mode	Guard condition	Result
By Count	Current Count > 0 and Playing	Spawn a balloon and renew the timer
By Count	Current Count ≤ 0 (attempts exhausted)	No spawn; the chain terminates
By Time	Current Time > 3.0 and Playing	Spawn a balloon and renew the timer
By Time	Current Time ≤ 3.0 (final 3 seconds)	No spawn; the chain terminates
Any mode	Game State ≠ Playing (game over)	No spawn; the chain terminates

Blueprint screenshots





Key points

- The dual check decides spawn start/stop: the mode (attempts/time) answers "should this mode still spawn", and the state (Playing) answers "is the game still running"; if either fails, no spawn and no timer renewal this round — the chain terminates naturally
- ByTime's 3-second threshold: spawning stops for the last 3 seconds, because a balloon spawned in the final second would not have risen before the game ends — meaningless
- Size and interval share the 0-value switch convention, echoing SetLevelData's configuration (Balloon Size=0 goes random size, Fixed Spawn Time=0 goes random time)
- Looping unchecked: the timer fires once and the event self-schedules its renewal, so the interval can be re-randomized each time

2.7 Select Spawn Position — Choosing the Spawn Position

Purpose

Returns one spawn point's world position per Is Random Pos?, supporting both random selection and sequential rotation, and picks only among the first Use Pos Num spawn points enabled by the current difficulty.

Blueprint screenshots

